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INFORMATION PROCESSING DEVICE, MANAGEMENT METHOD, AND NON-TRANSITORY COMPUTER-READABLE STORAGE MEDIUM STORING PROGRAM

Abstract

An information processing device is an information processing device for managing an image forming device. The information processing device includes: a storage unit configured to store first authentication information for accessing the image forming device and second authentication information that is information obtained by updating the first authentication information; and an acquisition unit configured to acquire device information of the image forming device by accessing the image forming device, in which the acquisition unit acquires, when acquisition of the device information of the image forming device by access using the second authentication information fails, the device information of the image forming device using the first authentication information.

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Background/Summary

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-026252, filed Feb. 26, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to an information processing device, a management method, and a non-transitory computer-readable storage medium storing a program.

2. Related Art

[0003] Simple Network Management Protocol (SNMP) is known as a standard protocol for managing devices in a Transmission Control Protocol (TCP)/Internet Protocol (IP) network environment. By using the SNMP, for example, a management device can acquire device information of a device to be managed. In the SNMP, in order to collect device information on the device to be managed, the management device needs to use authentication information set in the device to be managed. In general, in order to operate a system more securely, it is important to update authentication information required to access a device. In this regard, for example, JP-A-2003-288189 discloses a technique of periodically updating authentication information for using a printer connected to a network.

[0004] Authentication information set in each of the devices to be managed in order for the management device to manage the device to be managed may also be updated periodically or irregularly. When new authentication information is set in the device to be managed, the management device can access the device to be managed by using the new authentication information. However, for some reason, new authentication information, that is, updated authentication information may not be set in the device to be managed. In this case, the management device may not be able to access the device to be managed.

SUMMARY

[0005] An information processing device according to the present disclosure is an information processing device for managing an image forming device. The information processing device includes: a storage unit configured to store first authentication information for accessing the image forming device and second authentication information that is information obtained by updating the first authentication information; and an acquisition unit configured to acquire device information of the image forming device by accessing the image forming device, in which the acquisition unit acquires, when acquisition of the device information of the image forming device by access using the second authentication information fails, the device information of the image forming device using the first authentication information.

[0006] A management method according to the present disclosure is a management method for managing an image forming device. The management method includes: storing, in a storage unit in advance, first authentication information for accessing the image forming device and second authentication information that is information obtained by updating the first authentication information; and acquiring device information of the image forming device using the first authentication information when acquisition of the device information of the image forming device

by access using the second authentication information fails.

[0007] A non-transitory computer-readable storage medium storing a program according to the present disclosure causes a computer of an information processing device for managing an image forming device to execute operations including: a storage processing step of storing, in a storage unit, first authentication information for accessing the image forming device and second authentication information that is information obtained by updating the first authentication information; and an acquisition step of acquiring device information of the image forming device by accessing the image forming device, in which in the acquisition step, when acquisition of the device information of the image forming device by access using the second authentication information fails, the device information of the image forming device is acquired using the first authentication information.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a diagram showing a configuration of a management system according to an embodiment.

[0009] FIG. 2 is a block diagram showing an example of the functional configuration of an information processing device according to the embodiment.

[0010] FIG. 3 is a diagram showing an example of information stored in a setting DB.

[0011] FIG. 4 is a diagram showing an example of information stored in a history information DB.

[0012] FIG. 5 is a diagram showing an example of information displayed on a UI device.

[0013] FIG. 6 is a block diagram showing an example of a hardware configuration of the information processing device according to the embodiment.

[0014] FIG. 7 is a flowchart showing a flow of processing related to acquisition of device information.

DESCRIPTION OF EMBODIMENTS

[0015] An embodiment will be described below with reference to the drawings. To clarify the description, omission and simplification have been made as appropriate in the description and the drawings below. In the drawings, the same elements are denoted by the same reference signs, and the redundant description thereof is omitted as necessary.

[0016] FIG. 1 is a diagram showing a configuration of a management system **1** according to the embodiment. The management system **1** includes an information processing device **100** as a management device that manages an image forming device **200**, and the image forming device **200** managed by the information processing device **100**. The information processing device **100** and the image forming device **200** are connected to a network N such as the Internet.

[0017] The image forming device **200** is, for example, a device having a printing function, that is, a printer. The image forming device **200** may not necessarily have the printing function. The image forming device **200** may have one or more other functions instead of the printing function or in addition to the printing function. For example, the image forming device **200** may have a scanner function. In the example shown in FIG. 1, three image forming devices **200** are shown as the image forming device managed by the information processing device **100**, but the number of image forming devices **200** managed by the information processing device **100** is not limited to three and may be any number. The image forming devices **200** managed by the information processing device **100** may be image forming devices of the same model, or may be image forming devices of different models.

[0018] As described above, the information processing device **100** is a device for managing the image forming device **200**. The information processing device **100** may be referred to as a management device. The information processing device **100** manages the image forming device

200 by acquiring device information on the image forming device **200** using SNMP. Here, the device information is information indicating a state of the image forming device **200**. The device information may be, for example, a state indicating a state related to a printing function of the image forming device **200**. Specifically, the device information may be information indicating a remaining number of consumables (for example, ink) used for printing, or may be information indicating an error (for example, paper jam) occurring in the image forming device **200**. These are merely examples of the device information, and any device information may be acquired.

[0019] In order to achieve communication according to the SNMP, the information processing device **100** and the image forming device **200** need to use the same authentication information. More specifically, the information processing device **100** needs to access the image forming device **200** using the same authentication information as authentication information set in the image forming device **200**. Here, the authentication information is information used in authentication processing executed by the image forming device **200** to confirm whether processing of acquiring the device information of the image forming device **200** by the information processing device **100** is processing executed by a device having legitimate authority. The authentication information can be said to be information necessary for the information processing device **100** to access the image forming device **200**. Only when the information processing device **100** accesses the image forming device **200** using the same authentication information as the authentication information set in the image forming device **200**, the image forming device **200** transmits device information to the information processing device **100** assuming that the processing is executed by a device having legitimate authority.

[0020] In the embodiment, as an example, the information processing device **100** and the image forming device **200** use SNMP version v3. Therefore, in the embodiment, the authentication information includes the following information according to a specification of the SNMP version v3. That is, in the embodiment, the authentication information includes information designating a user name, information designating an authentication algorithm, information indicating an authentication password, information designating an encryption algorithm, information indicating an encryption password, and a context name. Other versions of SNMP may also be used. For example, in SNMP version v1 or version v2c, when accessing the image forming device **200**, the information processing device **100** needs to transmit an SNMP community name set in the image forming device **200** to the image forming device **200**. Only when the information processing device **100** uses an appropriate SNMP community name, access to the image forming device **200** is successful. Therefore, in the version v1 or the version v2c, the SNMP community name required for access can be said to be a type of authentication information.

[0021] In order to securely operate the management system **1**, the authentication information set in the image forming device **200** is updated periodically or irregularly by an update operation by an administrator. In the embodiment, in order to facilitate management of the authentication information set in the image forming device **200** and the update operation of the authentication information, the authentication information is set in each image forming device **200** so that all image forming devices **200** in the management system **1** use the same authentication information. Accordingly, when the update operation of the authentication information is performed, the authentication information common to all the image forming devices **200** is set in each image forming device **200**. Therefore, in the embodiment, it can be said that the information processing device **100** is a device that acquires the device information of each of the image forming devices **200** by accessing each of the image forming devices **200** using the authentication information set in common to the plurality of image forming devices **200**.

[0022] In this way, in the embodiment, the management system **1** is a system having the following features. Each of the plurality of image forming devices **200** is operated so that common authentication information according to SNMPv3 is set. The information processing device **100** holds the common authentication information, accesses each image forming device **200** using the

common authentication information, and acquires device information from each image forming device **200**.

[0023] The system having the above-described features has the following problems. When common authentication information (first authentication information) set in each image forming device **200** is updated to new common authentication information (second authentication information), new authentication information may not be set in some of the image forming devices **200**. That is, in some of the image forming devices **200**, the second authentication information that is new authentication information may not be reflected, and the first authentication information may continue to be used. In such a case, when a device used as the management device changes the authentication information used for acquiring the device information from the image forming device **200** from the first authentication information to the second authentication information, the device cannot acquire the device information from the image forming device **200** in which the first authentication information remains set. Therefore, in the embodiment, the information processing device **100** having a feature to solve such a problem will be described. For example, in the following case, an event occurs in which the authentication information set in the image forming device **200** is not updated. When new authentication information is set, in a case where the image forming device **200** is powered off, the updated authentication information is not set in the image forming device **200**. When an error such as a packet loss occurs during communication for setting new authentication information, updating of the authentication information in the image forming device **200** fails. When the number of image forming devices **200** is large, it is necessary to perform an authentication information update operation for the image forming devices **200** over a plurality of schedules, and thus old authentication information is continuously used in the image forming device **200** for which the update operation is not completed.

[0024] FIG. **2** is a block diagram showing an example of a functional configuration of the information processing device **100**. As shown in FIG. **2**, the information processing device **100** includes a setting unit **110**, a storage unit **120**, a storage processing unit **130**, an acquisition unit **140**, and a UI processing unit **150**.

[0025] The setting unit **110** sets authentication information common to all the image forming devices **200** in the management system **1** to each of the image forming devices **200**. That is, the setting unit **110** sets new authentication information in the image forming device **200** or updates the authentication information set in the image forming device **200**. The setting unit **110** sets, for example, new authentication information designated by a user via a UI device **161** described later in the image forming device **200**. The setting unit **110** accesses the image forming device **200** using authentication information currently set in the image forming device **200**, thereby setting updated authentication information. As described above, the updating of the authentication information is not always successful at the same timing in all the image forming devices **200**.

[0026] The storage unit **120** is a storage area for storing information used by the information processing device **100** to acquire device information from the image forming device **200**. In the embodiment, the storage unit **120** includes a setting DB (database) **121** and a history information DB (database) **122**.

[0027] The setting DB **121** is an area in which authentication information set in each image forming device **200** by the setting unit **110** as authentication information common to all the image forming devices **200** is stored. Every time the setting unit **110** sets new authentication information in the image forming device **200**, the new authentication information is added to the setting DB **121**. Therefore, the setting DB **121** stores not only the latest authentication information but also past authentication information. The setting DB **121** is a specific example of a storage unit that stores the first authentication information for accessing the image forming device **200** and the second authentication information that is information obtained by updating the first authentication information.

[0028] The history information DB **122** is an area in which information indicating authentication

information used to acquire device information of the image forming device **200** is stored in association with identification information of the image forming device **200**. That is, the history information DB **122** is an area in which information indicating authentication information for which device information of the image forming device **200** is successfully acquired and identification information of the image forming device **200** are stored in association with each other. More specifically, in the history information DB **122**, only information indicating the authentication information used by the acquisition unit **140** when the acquisition of the device information of the image forming device **200** is successful most recently is stored in association with the identification information of the image forming device **200**. Therefore, in the history information DB **122**, the number of pieces of information indicating the authentication information stored in association with the identification information of one image forming device **200** is one or less. For example, the information indicating the authentication information may be information for specifying one piece of authentication information from among various pieces of authentication information, and may be, for example, identification information for identifying the authentication information, or the authentication information itself.

[0029] The storage processing unit **130** stores information in the storage unit **120**. In the embodiment, every time the setting unit **110** sets new authentication information in the image forming device **200**, the storage processing unit **130** adds new authentication information to the setting DB **121**. Therefore, the setting DB **121** stores the updated authentication information every time the authentication information is updated. In other words, every time new authentication information is introduced into the management system **1**, the setting DB **121** stores newly introduced authentication information as updated authentication information. In the embodiment, the storage processing unit **130** stores the authentication information together the information indicating a priority order in the setting DB **121**. The priority order indicates which authentication information stored in the setting DB **121** is to be preferentially used when the authentication information stored in the setting DB **121** is used. In the embodiment, specifically, the priority order is an order according to an order in which the authentication information is updated. More specifically, the priority order is set for each piece of authentication information such that newer authentication information has a higher priority, that is, is used with higher priority.

[0030] In the embodiment, information indicating a priority order is automatically assigned to the authentication information stored in the setting DB **121** by processing of the storage processing unit **130**, but information indicating a priority order may be set for each piece of the authentication information stored in the setting DB **121** by the administrator using the information processing device **100**. In the embodiment, the storage processing unit **130** automatically adds new authentication information to the setting DB **121** every time the setting unit **110** sets the new authentication information in the image forming device **200**, but the authentication information may also be stored in the setting DB **121** by an operation of the administrator.

[0031] FIG. **3** is a diagram showing an example of information stored in the setting DB **121**. In the example shown in FIG. **3**, the setting DB **121** stores N different pieces of authentication information. As shown in FIG. **3**, a priority order is set for each piece of authentication information. In the example shown in FIG. **3**, the authentication information at a top of a table has a higher priority, and thus the authentication information at the top of the table indicates newer authentication information.

[0032] When the acquisition unit **140** acquires device information of the image forming device **200**, the storage processing unit **130** stores identification information of the image forming device **200** and information indicating the authentication information used to acquire the device information in association with each other as history information in the history information DB **122**. That is, when the acquisition unit **140** successfully acquires the device information, the storage processing unit **130** saves, in the history information DB **122**, identification information of the image forming device **200** and information indicating the authentication information used for

access at the time of acquiring the device information, in association with each other. In particular, when the acquisition unit **140** acquires the device information of the image forming device **200** using authentication information different from authentication information specified based on history information stored in the history information DB **122**, the storage processing unit **130** updates the history information with information indicating the authentication information used to acquire the device information. This will be described in more detail. In the present disclosure, among the plurality of image forming devices **200** managed by the information processing device **100**, one image forming device **200** whose device information is to be acquired by the acquisition unit **140** is referred to as a target device. For example, it is assumed that identification information of the image forming device **200** as the target device and information indicating authentication information B are stored in the history information DB **122** in association with each other. At this time, when the acquisition unit **140** successfully acquires the device information of the target device using authentication information A different from the authentication information B, the storage processing unit **130** changes the information associated with the identification information of the target device from the information indicating the authentication information B to information indicating the authentication information A. That is, the storage processing unit **130** updates information indicating the history information that is associated with the identification information of the target device. Accordingly, when the device information is acquired from the target device next time, the device information can be efficiently acquired using the history information.

[0033] FIG. **4** is a diagram showing an example of information stored in the history information DB **122**. As shown in FIG. **4**, the history information DB **122** saves, as history information, identification information of the image forming device **200** managed by the information processing device **100** and information indicating authentication information in association with each other. Only when the acquisition unit **140** successfully acquires the device information of the image forming device **200**, the history information of the image forming device **200** is saved in the history information DB **122**. Therefore, for the image forming device **200** for which there is no record of successfully acquiring the device information, there is no history information in the history information DB **122**.

[0034] The acquisition unit **140** accesses the image forming device **200** and acquires device information of the image forming device **200**. In the embodiment, in particular, the acquisition unit **140** accesses each of the plurality of image forming devices **200** and acquires the device information of the image forming device **200**. At this time, the acquisition unit **140** accesses the image forming device **200** using authentication information. When the authentication information used by the image forming device **200** matches authentication information set in the image forming device **200** as an access destination, acquisition of the device information from the image forming device **200** is successful. However, when the authentication information used by the image forming device **200** is different from the authentication information set in the image forming device **200** as the access destination, the acquisition of the device information from the image forming device **200** fails. For example, it is assumed that the acquisition unit **140** accesses each image forming device **200** using the updated authentication information (second authentication information) in a situation where the authentication information is not updated in some of the image forming devices **200** among all image forming devices **200** managed by the information processing device **100**. In this case, the acquisition unit **140** succeeds in acquiring the device information from the image forming device **200** in which the updated authentication information (second authentication information) is set. However, the acquisition unit **140** fails to acquire device information from the image forming device **200** in which the authentication information before update (first authentication information) is still set. Therefore, in the embodiment, when the acquisition of the device information of the image forming device **200** by access using the updated authentication information (second authentication information) fails, the acquisition unit **140** acquires the device information of the image forming device **200** using the authentication information before the update (first

authentication information). That is, after trying to acquire the device information of the image forming device **200** using the second authentication information, the acquisition unit **140** attempts to acquire the device information using the first authentication information for the image forming device **200** from which acquisition of the device information using the second authentication information fails. Accordingly, even in a situation in which authentication information is not updated in some of the image forming devices **200** managed by the information processing device **100**, it is possible to acquire device information from each image forming device **200**. In the embodiment, acquisition processing of the acquisition unit **140** described above is executed when history information about a target device (that is, one image forming device **200** from which an attempt is made to acquire device information) does not exist in the history information DB **122**. [0035] When the history information of the target device exists in the history information DB **122**, the acquisition unit **140** first attempts to acquire the device information of the target device using the authentication information specified based on the history information among the authentication information stored in the setting DB **121**. That is, when newly acquiring the device information of the image forming device **200**, the acquisition unit **140** specifies authentication information used to access the image forming device **200** based on the history information stored in the history information DB **122**. Accordingly, authentication information that has a record of successfully acquiring device information is used preferentially, and thus occurrence of failure in acquiring device information can be reduced. In other words, occurrence of unnecessary communication can be reduced.

[0036] When the acquisition of the device information of the target device using the authentication information specified based on the history information fails, the acquisition unit **140** acquires the device information of the target device using the authentication information selected according to a predetermined priority order among the authentication information stored in the setting DB **121**. More specifically, in this case, the acquisition unit **140** attempts to acquire the device information of the target device by using the authentication information stored in the setting DB **121** in order from authentication information having the highest priority. Accordingly, the acquisition unit **140** can find and use newly set authentication information in the target device. Therefore, even when acquisition of the device information using the authentication information specified based on the history information fails, the device information can be acquired.

[0037] The UI processing unit **150** executes processing related to a user interface. For example, the UI processing unit **150** executes processing of receiving an instruction or information input from a user (for example, an administrator) via the UI device **161** described later. The UI processing unit **150** executes processing of outputting information using the UI device **161** described later. FIG. 5 is a diagram showing an example of information displayed on the UI device **161** by the processing of the UI processing unit **150**. The UI processing unit **150** executes, for example, processing of displaying a list of device information of each image forming device **200** that is a device to be managed by the information processing device **100**. That is, the UI processing unit **150** displays the device information of the image forming device **200** acquired by the acquisition unit **140** on the UI device **161**. As shown in FIG. 5, the UI device **161** may indicate a priority order set in the authentication information used to acquire the device information by associating the authentication information used to acquire the device information with the identification information of the image forming device **200** so that the administrator can recognize the authentication information. The UI processing unit **150** can also indicate any information in association with the identification information of the image forming device **200**. For example, as shown in FIG. 5, the UI processing unit **150** may indicate a date and time the device information is acquired. When there is an image forming device **200** from which the device information cannot be acquired, the UI processing unit **150** may display information indicating an error in acquisition of the device information instead of displaying the device information of the image forming device **200**.

[0038] The UI processing unit **150** can display not only device information but also any

information on the UI device **161**. For example, the UI processing unit **150** may display, on the UI device **161**, a UI screen on which a user sets a priority order for each piece of authentication information stored in the setting DB **121**.

[0039] Next, a hardware configuration of the information processing device **100** will be described. FIG. **6** is a block diagram showing an example of the hardware configuration of the information processing device **100**. As shown in FIG. **6**, the information processing device **100** includes the UI device **161**, a network interface **162**, a memory **163**, and a processor **164**.

[0040] The UI device **161** is a device functioning as a user interface. Specifically, the UI device **161** is implemented by a display that displays various kinds of information and an input device such as a keyboard or a pointing device that receives an input operation from a user. The UI device may be implemented as a touch panel in which a display and an input device are integrated together.

[0041] The network interface **162** is used to perform communication via the network N. The network interface **162** may include, for example, a network interface card (NIC).

[0042] The memory **163** is implemented by, for example, a combination of a volatile memory and a nonvolatile memory. The memory **163** is used to store a program or the like executed by the processor **164**. The storage unit **120** described above may be implemented by a storage device such as the memory **163**. The memory **163** may include a plurality of memories.

[0043] The processor **164** reads a program from the memory **163** and executes the program. Accordingly, the processor **164** implements functions of the setting unit **110**, the storage processing unit **130**, the acquisition unit **140**, and the UI processing unit **150** described above. The processor **164** may be, for example, a microprocessor, a microprocessor unit (MPU), or a central processing unit (CPU). The processor **164** may include a plurality of processors. In this way, the information processing device **100** has functions as a computer.

[0044] The program includes an instruction group (or software codes) for causing a computer to perform one or more of the functions explained in the embodiment when the program is read in the computer. The program may be stored in a non-transitory computer-readable medium or a tangible storage medium. Not as a limitation but as an example, the computer-readable medium or the tangible storage medium includes a random-access memory (RAM), a read-only memory (ROM), a flash memory, a solid-state drive (SSD), or devices based on other memory technologies, a CD-ROM, a digital versatile disc (DVD), a Blu-ray (trademark registered) disc, or other optical disc storages, a magnetic cassette, a magnetic tape, a magnetic disk storage or other magnetic storage devices. The program may be transmitted on a transitory computer-readable medium or a communication medium. Not as a limitation but as an example, the transitory computer-readable medium or the communication medium includes an electric, optical, or acoustic propagation signal or a propagation signal of another form.

[0045] Next, a flow of processing of the information processing device **100** related to acquisition of device information will be described. FIG. **7** is a flowchart showing a flow of processing of the information processing device **100** related to acquisition of device information. Hereinafter, the flow of processing will be described with reference to FIG. **7**.

[0046] In step **S100**, the acquisition unit **140** determines whether history information about a target device exists in the history information DB **122**. When the history information about the target device does not exist in the history information DB **122**, the processing proceeds to step **S101**. For example, when the device information of the target device is acquired for the first time, the processing proceeds to step **S101**. In contrast, when the history information about the target device exists in the history information DB **122**, the processing proceeds to step **S107**. For example, when there is a record of successfully acquiring the device information of the target device, the processing proceeds to step **S107**.

[0047] In step **S101**, the acquisition unit **140** executes acquisition processing of acquiring the device information of the target device using authentication information having the highest priority

among authentication information stored in the setting DB **121**. That is, the acquisition unit **140** attempts to acquire the device information from the target device by accessing the target device using the authentication information having the highest priority. In the acquisition processing, when the device information of the target device is successfully acquired (YES in step **S102**), the processing proceeds to step **S106**. In contrast, when the acquisition of the device information of the target device fails (NO in step **S102**), the processing proceeds to step **S103**.

[0048] In step **S103**, the acquisition unit **140** determines whether authentication information having the next highest priority exists in the setting DB **121**. When the authentication information having the next highest priority does not exist in the setting DB **121** (NO in step **S103**), that is, when the processing of acquiring the device information is attempted for all authentication information stored in the setting DB **121**, the processing ends. In contrast, when the authentication information having the next highest priority exists in the setting DB **121**, the processing proceeds to step **S104**.

[0049] In step **S104**, the acquisition unit **140** executes acquisition processing of acquiring the device information of the target device using the authentication information having the next highest priority among the authentication information stored in the setting DB **121**. That is, the acquisition unit **140** attempts to acquire the device information from the target device by accessing the target device using the authentication information determined to exist in the setting DB **121** in step **S103**. In the acquisition processing, when the device information of the target device is successfully acquired (YES in step **S105**), the processing proceeds to step **S106**. In contrast, when the acquisition of the device information of the target device fails (NO in step **S105**), the processing returns to step **S103**, and the above-described processing is repeated.

[0050] When the device information of the target device is successfully acquired, the storage processing unit **130** saves the history information in the history information DB **122** in step **S106**. That is, the storage processing unit **130** saves, in the history information DB **122**, the identification information of the target device and information indicating the authentication information for which the device information is successfully acquired, in association with each other. When the history information about the target device already exists in the history information DB **122**, the storage processing unit **130** updates the history information saved in the history information DB **122**. For example, when the processing proceeds to this step after it is determined in step **S100** that history information about the target device exists in the history information DB **122**, the history information is updated.

[0051] In step **S100**, when it is determined that the history information about the target device exists in the history information DB **122**, the acquisition unit **140** executes processing of acquiring the device information of the target device using authentication information specified based on the history information of the target device saved in the history information DB **122**. That is, when the processing proceeds to step **S107**, the acquisition unit **140** attempts to acquire the device information from the target device by accessing the target device using the authentication information specified based on the history information. In the acquisition processing, when the device information of the target device is successfully acquired (YES in step **S108**), the processing proceeds to step **S106** described above. That is, the history information of the history information DB **122** is updated for the target device. In contrast, when the acquisition of the device information of the target device fails (NO in step **S108**), the processing proceeds to step **S101** described above. That is, in this case, the acquisition unit **140** attempts to acquire the device information of the target device by sequentially using the authentication information saved in the setting DB **121** according to a priority order. In this case, use of the authentication information for which acquisition of the device information fails, that is, the authentication information specified based on the history information of the history information DB **122**, may be skipped in the processing of step **S101** and the subsequent processing.

[0052] The embodiment is described above. As described above, the information processing device **100** stores, in the storage unit **120** in advance, the first authentication information for accessing the

image forming device **200** and the second authentication information that is information obtained by updating the first authentication information. When the information processing device **100** fails to acquire the device information of the image forming device **200** by access using the second authentication information, the information processing device **100** acquires the device information of the image forming device **200** using the first authentication information. Therefore, even when new authentication information cannot be set in the device to be managed for some reason, the information processing device **100** can access the device to be managed and appropriately acquire the device information.

[0053] While the embodiment has been described above, the present disclosure is not limited to the above embodiment and changes can be made without departing from the spirit and scope of the present disclosure. For example, although the device managed by the information processing device **100** is the image forming device in the above-described embodiment, the device managed by the information processing device **100** is not limited to the image forming device and may be any electronic device. More specifically, the device managed by the information processing device **100** may be an electronic device capable of communicating with the information processing device **100** according to a predetermined protocol such as the SNMP.

Claims

1. An information processing device for managing an image forming device, the information processing device comprising: a storage unit configured to store first authentication information for accessing the image forming device and second authentication information that is information obtained by updating the first authentication information; and an acquisition unit configured to acquire device information of the image forming device by accessing the image forming device, wherein the acquisition unit acquires, when acquisition of the device information of the image forming device by access using the second authentication information fails, the device information of the image forming device using the first authentication information.
2. The information processing device according to claim 1, further comprising: a storage processing unit configured to, when the device information of the image forming device is acquired, store identification information of the image forming device and information indicating authentication information used to acquire the device information in association with each other as history information in the storage unit, wherein when the device information of the image forming device is newly acquired, the acquisition unit specifies authentication information used to access the image forming device based on the stored history information.
3. The information processing device according to claim 2, wherein when acquisition of the device information of the image forming device using the authentication information specified based on the history information fails, the acquisition unit acquires the device information using authentication information selected according to a predetermined priority order among authentication information stored in the storage unit.
4. The information processing device according to claim 3, wherein the priority order is an order according to an order in which authentication information is updated.
5. The information processing device according to claim 3, wherein when the device information of the image forming device is acquired using authentication information different from the authentication information specified based on the history information, the storage processing unit updates the history information with information indicating the authentication information used to acquire the device information.
6. A management method for managing an image forming device, the management method comprising: storing, in a storage unit in advance, first authentication information for accessing the image forming device and second authentication information that is information obtained by updating the first authentication information; and acquiring device information of the image

forming device using the first authentication information when acquisition of the device information of the image forming device by access using the second authentication information fails.

7. A non-transitory computer-readable storage medium storing a program, the program causing a computer of an information processing device for managing an image forming device to execute operations comprising: a storage processing step of storing, in a storage unit, first authentication information for accessing the image forming device and second authentication information that is information obtained by updating the first authentication information; and an acquisition step of acquiring device information of the image forming device by accessing the image forming device, wherein in the acquisition step, when acquisition of the device information of the image forming device by access using the second authentication information fails, the device information of the image forming device is acquired using the first authentication information.
